Bureau
International des
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Mesures

Le système international d'unités
The International System of Units

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Annexe 2
Annex 2

Valeurs recommandées des fréquences étalons destinées à la mise en pratique de la définition du mètre et aux représentations secondaires de la seconde

Recommended values of standard frequencies for applications including the practical realization of the metre and secondary representations of the second

Partie 2
Part 2

Iodine ()

Iodine ()

CCL-CCTF Frequency Standards Working Group



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Part 2: Iodine ()

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Appendix 2

1. CIPM recommended values

The values $f = 551\ 580\ 162\ 400\ \text{kHz}$
$\lambda = 543\ 515\ 663.608\ \text{fm}$

with a relative standard uncertainty of 4.5×10^{-11} apply to the radiation of a laser stabilized to an external iodine cell and subject to the following conditions:

- cold point temperature (0 ± 2) °C[(1)];
- frequency modulation width, peak-to-peak, (2.0 ± 0.5) MHz.

[(1)] For the specification of operating conditions, such as temperature, modulation width and laser power, the symbols ± refer to a tolerance, not an uncertainty.

2. Source data

Adopted value

$f = 551\ 580\ 162\ 400\ \ (25)$ kHz
$u_c/y = 4.5 \times 10^{-11}$

for which:
$\lambda = 543\ 515\ 663.608\ (24)$ fm
$u_c/y = 4.5 \times 10^{-11}$

calculated from

f/ kHz	u_c/y	source data
551 580 162 397.1	1.3×10^{-11}	[1]

Given the small number of calibrations and the individual behaviour of participating lasers in this calibration, the CCL considered it prudent to adopt an uncertainty of 25 kHz for the recommended value. The final CCL value chosen was $f = 551\ 580\ 162\ 400\ (25)$ kHz with the following operating conditions:

- cold point temperature (0 ± 2) °C[(1)];
- frequency modulation width, peak-to-peak, (2.0 ± 0.5) MHz.

It was felt appropriate not to define the laser type.

[(1)] For the specification of operating conditions, such as temperature, modulation width and laser power, the symbols ± refer to a tolerance, not an uncertainty.

3. Absolute frequency of the other transitions related to those adopted as recommended and frequency intervals between transitions and hyperfine components

These tables replace those published in BIPM Com. Cons. Long., 2001, **10**, 182-183 and *Metrologia*, 2003, **40**, 126.

The notation for the transitions and the components is that used in the source references. The values adopted for the frequency intervals are the weighted means of the values given in the references.

For the uncertainties, account has been taken of:

- the uncertainties given by the authors;
- the spread in the different determinations of a single component;
- the effect of any perturbing components;
- the difference between the calculated and the measured values.

In the tables, uc represents the estimated combined standard uncertainty (1σ).

All transitions in molecular iodine refer to the B-X system.

Table 1

$\lambda \approx 543$ nm $^{127}I_2$ R(12) 26-0

a_n	$[f(a_n) - f(b_{10})]$/MHz	u_c/ MHz	a_n	$[f(a_n) - f(b_{10})]$/MHz	u_c/ MHz
a_1	-1 162.24	0.02	a_9	-679.420	0.005
a_2	-909.87	0.02	a_{10}	-596.134	0.005
a_3	-900.11	0.03	a_{11}	-485.61	0.01
a_4	-853.336	0.005	a_{12}	-476.35	0.01
a_5	-848.131	0.005	a_{13}	-423.23	0.01
a_6	-795.92	0.01	a_{14}	-410.01	0.01
a_7	-752.382	0.005	a_{15}	-305.910	0.005
a_8	-733.134	0.005			

Frequency referenced to b_{10}, R(106) 28-0, $^{127}I_2$: $f = 551\,580\,162\,400$ kHz [2]

Ref. [3], [4], [5], [6], [7], [8], [9]

Table 2

$\lambda \approx 543$ nm $^{127}I_2$ R(106) 28-0

b_n	$[f(b_n) - f(b_{10})]$/MHz	u_c/ MHz	b_n	$[f(b_n) - f(b_{10})]$/MHz	u_c/ MHz
b_1	-573.765	0.005	b_9	-114.575	0.005

b_2	-320.462	0.005	b_{10}	0	–
b_3	-291.59	0.01	b_{11}	124.83	0.01
b_4	-282.143	0.005	b_{12}	132.31	0.01
b_5	-253.675	0.005	b_{13}	154.51	0.01
b_6	-172.693	0.005	b_{14}	162.65	0.01
b_7	-159.428	0.005	b_{15}	287.24	0.01
b_8	-127.760	0.005			
Frequency referenced to	b_{10}, R(106) 28-0, $^{127}I_2$: $f = 551\,580\,162\,400$ kHz [2]				
Ref. [3], [4], [5], [6], [7], [8], [9]					

Absorbing molecule $^{127}I_2$, b_{10} component, R(106) 28-0 transition[1]

[1] All transitions in I_2 refer to the $B^3\Pi\, 0_u^+ - X^1\, \Sigma_g^+$ system.

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